

Assignment 2 — Forecasting (TES) and Linear Programming

Individual • 100 points • Due Mon, Oct 5, 12:00 PM (noon) on Canvas

Submission instructions. Submit one PDF on Canvas, typed or neatly handwritten and scanned. Show your full problem-solving process; final answers alone will not receive full credit. Work every calculation yourself; a calculator or a spreadsheet used as a calculator is fine, but solver output and AI tools used to obtain the solutions directly (such as ChatGPT, Claude, or Copilot) are not allowed, and every step must be shown. The deadline is Monday, Oct 5, 12:00 PM (noon), three hours before the review session; the solutions are posted at noon and reviewed in the session, so late submissions are not accepted or lose points.

What this assignment covers. Problem 1 is triple exponential smoothing from Lecture 1. Problems 2–5 are linear programming from Lecture 2: formulating a model from a word problem, the graphical method with iso-profit lines, the corner-point method, binary variables, and a formulation in which downtime and defect rates enter the constraints. The software part of Lecture 2 (slides 41–49) is not in this assignment; it becomes part of the midterm project after fall break.

Problem 1 — Triple exponential smoothing (25 points)

A distributor of HVAC compressors sees strong quarterly seasonality. Quarterly sales (thousands of units) for the first year and the first quarter of the second year are:

Quarter	Q1 (t = 1)	Q2 (t = 2)	Q3 (t = 3)	Q4 (t = 4)	Q1, year 2 (t = 5)
Sales A_t	40	50	60	50	44

Use TES with $\alpha = 0.2$, $\beta = 0.2$, $\gamma = 0.2$ and season length $N = 4$, initialized on the first year exactly as in the lecture slides (the Notation box below gives the initialization). Keep four decimals in intermediate values.

- Initialization (6 pts).** Compute the level F_4 , the trend T_4 , and the four seasonal factors c_1, c_2, c_3, c_4 .
- First forecast (4 pts).** Compute the forecast f_5 made at the end of $t = 4$, and its error $\varepsilon_5 = f_5 - A_5$ once $A_5 = 44$ is observed.
- Update (10 pts).** Using $A_5 = 44$, compute F_5 , T_5 , and c_5 . Show each equation with the numbers substituted.
- Forecasts (5 pts).** Compute the one-step forecast f_6 and the two-step forecast f_7 , both made at the end of $t = 5$. State which seasonal factor each forecast uses and why.

Notation (as in the lecture slides). F_t level, T_t trend, c_t seasonal factor at the end of period t ; $N = 4$.
 $F_t = \alpha A_t / c_{t-N} + (1 - \alpha)(F_{t-1} + T_{t-1})$; $T_t = \beta(F_t - F_{t-1}) + (1 - \beta)T_{t-1}$; $c_t = \gamma A_t / F_t + (1 - \gamma)c_{t-N}$; $f(t, \tau) = (F_t + \tau T_t) \cdot c_{(t, \tau-N)}$ for $\tau = 1, \dots, N$. Initialization on the first season: $F_N = (A_1 + \dots + A_N)/N$, $T_N = 0$, $c_i = A_i / F_N$ for $i = 1, \dots, N$.

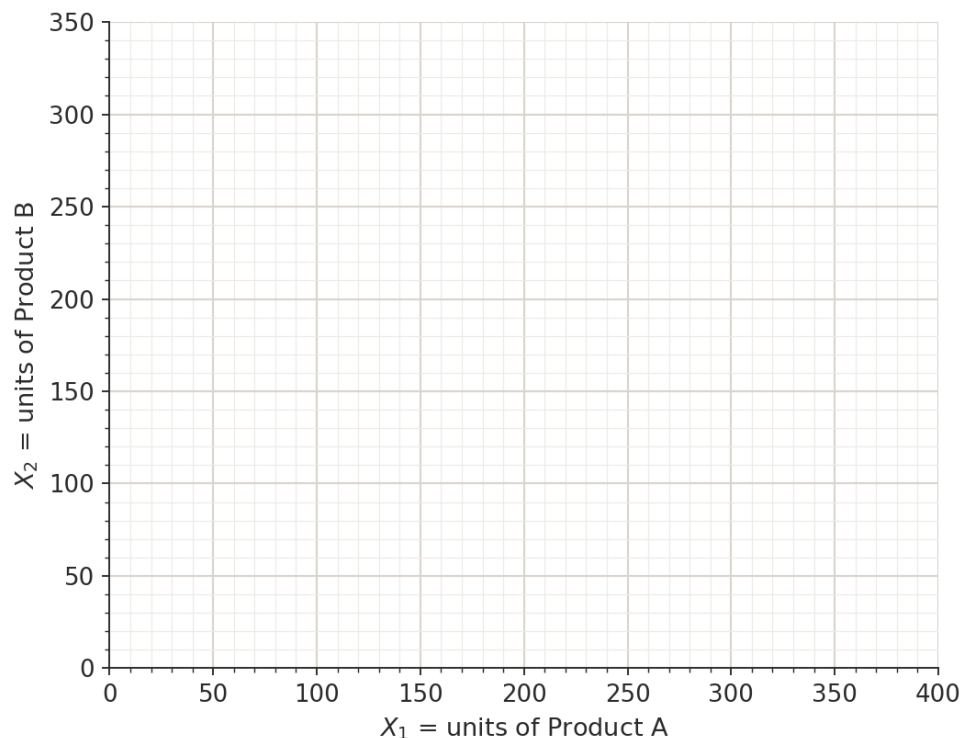
Problem 2 — Product mix by the graphical method (25 points)

A manager must decide the mix of products to produce for the coming week. Product A requires 3 minutes per unit for molding, 2 minutes per unit for painting, and 2 minutes per unit for packing.

Product B requires 2 minutes per unit for molding, 4 minutes per unit for painting, and 3 minutes per unit for packing. There are 600 minutes available for molding, 720 minutes for painting, and 480 minutes for packing. Both products have a contribution margin of \$1.50 per unit.

Resource	Product A (min/unit)	Product B (min/unit)	Minutes available
Molding	3	2	600
Painting	2	4	720
Packing	2	3	480
Contribution margin	\$1.50 per unit	\$1.50 per unit	

- (a) **Formulation (8 pts).** Define the decision variables and state the objective function and all constraints algebraically.
- (b) **Feasible region (9 pts).** Plot each constraint line on the grid below (or on your own graph), shade the feasible region, and label its corner points with their coordinates.
- (c) **Iso-profit line (8 pts).** Draw one iso-profit line for a profit value of your choice (state the value), then move it parallel to itself to find the optimal product mix and the maximum weekly contribution. Which resource is never binding (a constraint is binding when the optimal point lies on its line, that is, the resource is used to its limit; otherwise it has slack), and what does that mean for the manager?



Problem 3 — Rienzi Farms by the corner-point method (15 points)

Rienzi Farms grows sugar cane and soybeans on its 500 acres of land. An acre of soybeans yields a profit contribution of \$1,000, while an acre of sugar cane contributes \$2,500. Because of a government program, no more than 150 acres may be planted with soybeans. During the planting season, 1,800 hours of planting time are available. Each acre of soybeans requires 2 hours to plant, whereas each acre of sugar cane requires 6 hours. The company seeks to maximize its total contribution (profit) from the planting decision.

- (a) **Formulation (6 pts).** Formulate the problem as a linear program (decision variables, objective, constraints).
- (b) **Corner-point method (9 pts).** Sketch the feasible region. List every corner point together with the two constraints that define it, evaluate the objective at each corner, and report the optimal plan and its profit. How many acres are left unplanted at the optimum?

Problem 4 — Rollins Publishing: which books to publish (20 points)

Rollins Publishing must decide which textbooks from the following table to publish. For each book, the maximum demand, fixed publishing cost, variable cost per copy, and selling price per copy are given.

Book	Max demand (D_i)	Fixed cost (F_i)	Variable cost (V_i)	Selling price (S_i)
1	9,000	\$12,000	\$19	\$40
2	8,000	\$21,000	\$28	\$60
3	5,000	\$15,000	\$30	\$52
4	6,000	\$10,000	\$20	\$34
5	7,000	\$18,000	\$20	\$45

Rollins has the capacity to publish a total of 20,000 books.

- (a) **Formulation (16 pts).** Formulate a model to decide which books to publish and how many copies of each, to maximize total profit. Use a binary variable Y_i for the publish decision and a continuous variable X_i for the number of copies. State the objective, the capacity constraint, the constraints that link X_i to Y_i , and the variable domains. Formulation only; do not solve.
- (b) **Spot the error (4 pts).** A classmate writes the linking constraint as $X_i \geq D_i Y_i$. In two or three sentences, explain what this constraint would allow and why it is wrong.

Problem 5 — Machine shop with downtime and defects (15 points)

A machine shop processes two types of parts, both of which involve three consecutive steps: (1) polishing, (2) drilling, and (3) tapping. These steps are performed by three machine tools that have downtime rates of 5%, 10%, and 20%, respectively. In addition, the defective rates for the three steps are $\alpha = 1\%$, $\beta = 5\%$, and $\gamma = 5\%$, respectively; a part found defective at a step is scrapped and does not continue to the next step.

Product A requires 5 minutes of polishing, 2 minutes of drilling, and 2 minutes of tapping. Product B requires 2 minutes of polishing, 5 minutes of drilling, and 3 minutes of tapping. The total available work time for polishing, drilling, and tapping is 2,400 minutes per week for each process. The unit net profits of Products A and B are \$5 and \$3 per good (non-defective) unit, respectively.

Formulate a linear programming model to determine how many units of Products A and B should be started in one week to maximize profit. Define the decision variables, and show explicitly (i) how the downtime rates change the minutes available at each step, (ii) how the defect rates change the number of units that reach each step, and (iii) how they enter the objective. Formulation only; do not solve.

Hint. Two effects are at work. Downtime reduces the minutes a machine is actually available, so the right-hand side of each capacity constraint is the available time after downtime, not 2,400.

Defects act in sequence: a part scrapped at polishing never reaches drilling, and a part scrapped at drilling never reaches tapping. So if X units of a product are started, only $(1 - \alpha)X$ of them reach drilling, and $(1 - \alpha)(1 - \beta)X$ reach tapping; write each machine's time constraint on the units that actually reach that machine, and base the objective on the good units that come out of the last step.

Rubric summary

Problem	Points	Full credit looks like
1 TES	25	Correct initialization; f_5 and its error; F_5 , T_5 , c_5 with the equations shown; f_6 and f_7 with the right seasonal factors identified.
2 Graphical method	25	Correct model; all three lines drawn, region shaded, corners labeled; an iso-profit line moved to the optimum; the never-binding resource named and interpreted.
3 Corner-point method	15	Correct model; every corner with its defining constraints and objective value; the optimum and the unplanted acreage.
4 Binary variables	20	Objective with fixed costs on Y_i ; capacity; $X_i \leq D_i Y_i$ links; $Y_i \in \{0, 1\}$, $X_i \geq 0$; a clear explanation of what the wrong link allows.
5 Downtime and defects	15	Variables defined as units started; available minutes reduced by downtime; step-by-step yields in the constraints; objective on good units; nonnegativity.

Questions: office hours (ISTB12 355J) or Piazza.